Problem set #1

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1 Business Cycle Data

In this section, I analyze quarterly business cycle fluctuations in data on real Gross Domestic Product (GDP) and real Personal Consumption Expenditures (PCE) for the United States and Italy. To do so, I wrote a Python script that, through an API key, directly downloads the relevant time series from the Federal Reserve Economic Data (FRED) database (Federal Reserve Bank of St. Louis 2026).

After downloading the data, I take the natural logarithm of each series and extract their cyclical components using the Hodrick-Prescott (HP) filter (Hodrick and Prescott 1997) with the standard smoothing parameter for quarterly data, $\lambda = 1600$. Since the Italian series starts later than the US one, the US sample is truncated so that both countries are analysed over a common time interval, from 1996 Q1 to 2025 Q4.

The detrended series for GDP and PCE are plotted below.

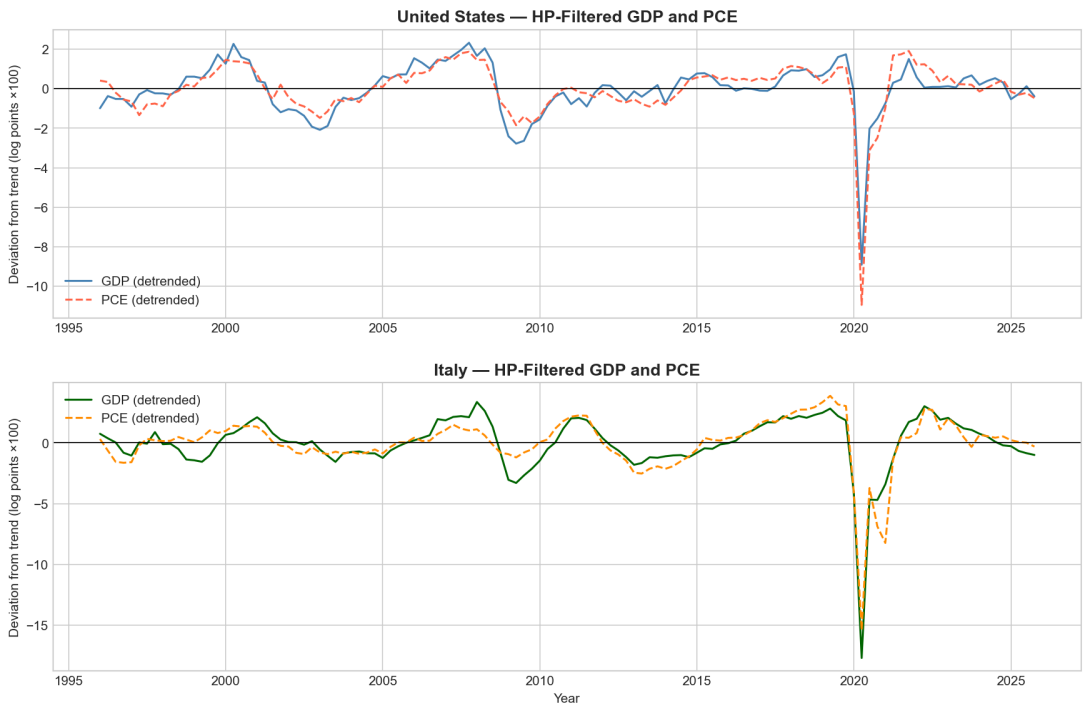

Figure 1: HP-filtered cyclical components of real GDP and real PCE for the United States and Italy

In addition, I compute a set of summary statistics for the detrended series, the variance of cyclical GDP and cyclical PCE for each country, the correlation between detrended GDP and detrended PCE within each country, and the correlation between Italy's detrended variables and US detrended GDP. Variances are computed using HP-filtered log deviations multiplied by 100, so the cyclical components are expressed approximately in percentage deviations from trend. The results are reported in Table 1.

Statistic	US	Italy
Var(GDP cycle)	1.7270	5.2202
Var(PCE cycle)	1.8870	4.9950
Corr(GDP, PCE)	0.9062	0.9035
Corr(Italy GDP, US GDP)	—	0.7581
Corr(Italy PCE, US GDP)	—	0.6686

Table 1: Summary statistics for HP-filtered GDP and PCE over the full sample period

As a result, we obtain that GDP and PCE cycles are positively correlated in both countries, and that Italy's GDP and PCE cycles are also positively correlated with US GDP. This is in line with prior expectations, since consumption is generally procyclical and international business cycles tend to display a significant common component, reflecting the role of global shocks and economic interdependence across countries. It is also worth noting that Italy displays higher volatility than the United States, consistent with the view that the Italian economy is more exposed to structural vulnerabilities and presents more fluctuations.

However, the variance results for the United States do not appear fully consistent with standard business cycle regularities, since the variance of detrended PCE is slightly larger than that of detrended GDP. This is somewhat unusual, as consumption is typically expected to be less volatile than output over the business cycle. A plausible explanation is that the COVID-19 period, together with the exceptional collapse and rebound that followed, has a substantial effect on the estimated second moments of the series. For this reason, it could be useful to repeat the analysis excluding the pandemic period and restricting the sample to observations up to 2019 Q4.

The corresponding summary statistics are reported in Table 2.

Statistic	US	Italy
Var(GDP cycle)	1.1027	1.5222
Var(PCE cycle)	0.6644	1.0164
Corr(GDP, PCE)	0.8797	0.7283
Corr(Italy GDP, US GDP)	—	0.5279
Corr(Italy PCE, US GDP)	—	0.4001

Table 2: Summary statistics for HP-filtered GDP and PCE excluding the COVID-19 period (sample ending in 2019 Q4)

Once the COVID-19 period is removed, the results look more consistent with conventional business cycle facts. In particular, detrended consumption becomes less volatile than detrended output in both countries, and GDP and PCE remain positively correlated.

2 Local Projections and Macroeconomic Shocks

2.1 Replication from Ramey 2016

In this section, I replicate one of the impulse response functions reported in (Ramey 2016) using Jordà's local projection method (Jordà 2005). More specifically, I reproduce the monetary policy shock exercise corresponding to Figure 3.2B, based on the original Romer and Romer monetary shock series (C. D. Romer and D. H. Romer 2004). The sample period in the database available on Ramey's website (Ramey 2026) is 1969 M3 to 1996 M12.

The shock variable is the original Romer and Romer monetary policy shock, while the main outcome variables are the federal funds rate, industrial production, the unemployment rate, and the consumer price index. Following Ramey's specification, I estimate a sequence of local projection regressions for horizons up to 48 months. The set of controls includes two lags of the Romer-Romer shock, the federal funds rate, log industrial production, the unemployment rate, log CPI, and the log commodity price index. In addition, in order to preserve the recursiveness assumption used in the benchmark specification, I include the contemporaneous values of industrial production, unemployment, CPI, and commodity prices whenever appropriate. Standard errors are computed using a Newey-West heteroskedasticity and autocorrelation consistent correction (Newey and West 1987), and I report 90 % confidence intervals.

Figure 2 presents the estimated impulse responses from my replication together with the corresponding figure from Ramey, so that the comparison is immediate.

Overall, the replicated impulse responses are qualitatively similar to those reported by Ramey. In particular, the responses display the same broad pattern: the federal funds rate increases after the shock, industrial production declines, unemployment rises, and prices fall gradually.

2.2 Impulse Responses to a Marginal Efficiency of Investment Shock

As a second empirical exercise, I consider a different shock series and estimate its dynamic effects using local projections. The empirical question I address is how real non-residential

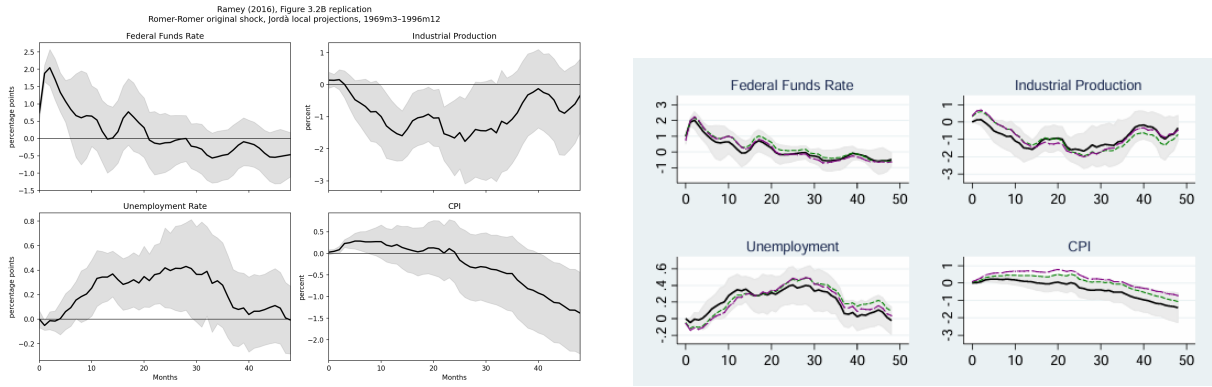

Figure 2: The left panel reports my estimated impulse response functions, the right panel reproduces the corresponding figure from Ramey for comparison. Shaded areas denote 90% confidence intervals

investment per capita responds to a positive marginal efficiency of investment (MEI) shock.

I use the MEI shock constructed by Justiniano, Primiceri, and Tambalotti (Justiniano, Primiceri, and Tambalotti 2011), available in the technology dataset distributed by Ramey (Ramey 2026). Economically, this disturbance captures innovations in the efficiency with which current investment expenditure is transformed into installed productive capital. In particular, Justiniano, Primiceri, and Tambalotti argue that shocks to the marginal efficiency of investment are an important driver of US business cycle fluctuations, especially for investment and other cyclical aggregates.

The outcome variable is real non-residential investment per capita, constructed as the logarithm of real non-residential investment divided by population. In the dataset, the sample runs from 1954 Q3 to 2009 Q1.

To estimate the impulse response, I use Jordà local projections with a horizon of 20 quarters. The shock is normalized by its sample standard deviation, so that the reported impulse response can be interpreted as the effect of a one-standard-deviation MEI shock. The specification includes a linear time trend and two lags of the shock, real GDP per capita, stock prices per capita, labor productivity, and the outcome variable itself. Formally, for each horizon h, I estimate

$$y_{t+h} = \alpha_h + \beta_h \varepsilon_t + \Gamma_h(L) X_t + \delta_h t + u_{t+h},$$

where y_{t+h} is log real non-residential investment per capita at horizon h, ε_t is the normalized MEI shock, and X_t collects the lagged controls. Inference is based on Newey-West heteroskedasticity and autocorrelation consistent standard errors, and I report 90% confidence intervals.

Figure 3 reports the estimated impulse response.

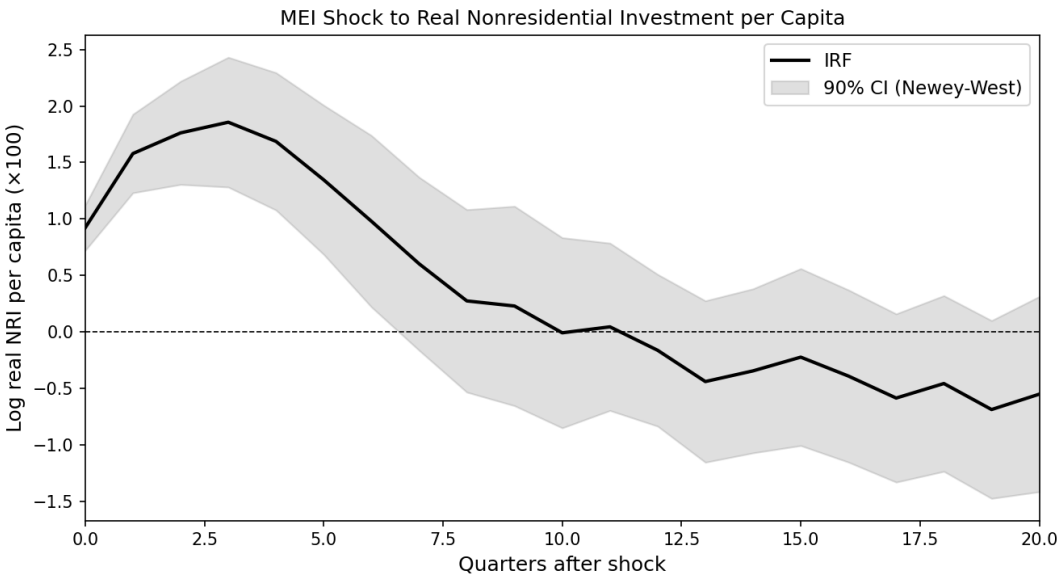

Figure 3: Impulse response of real non-residential investment per capita to a one-standard-deviation marginal efficiency of investment shock. The shaded area represents the 90% confidence interval

The estimated response is economically intuitive. The response peaks at around three to four quarters after the shock and subsequently declines, returning toward zero by approximately horizon twelve. Beyond that point, the impulse response becomes slightly negative, though the confidence intervals widen considerably, making it difficult to draw precise inference at longer horizons. A positive MEI shock increases the efficiency of investment and therefore raises the incentive to accumulate capital. As a result, real non-residential investment per capita responds positively after the shock. This pattern is consistent with the interpretation of MEI disturbances as shocks that improve the transformation of current resources into future productive capital and aligns with prior expectations and with the findings of Justiniano, Primiceri, and Tambalotti, who emphasize the central role of investment shocks in driving macroeconomic fluctuations.

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